STATISTICAL MECHANICS

Assignment 3 (Brownian motion Data Analysis)

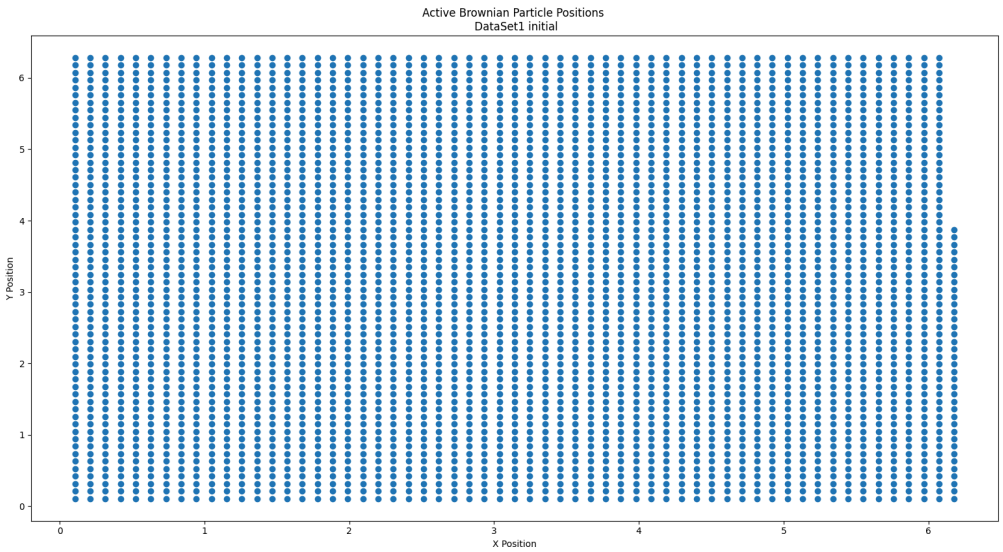
_____PART A_____

For data set 1

Initial Position

```
import numpy as np
import matplotlib.pyplot as plt

file_path = r'C:\Users\LC15\Desktop\Stat
Mech\Active_BrownianParticle_DataSet\Active_BrownianParticle_DataSet\Final
ized_data\DataSet1\part\POSITION0'
with open(file_path, 'rb') as myFile:
    data = np.fromfile(myFile, dtype=np.float64)

x_position = data[0:3517]
y_position = data[3517:7035]

print(data)
print(len(data))
plt.scatter(x_position, y_position)
plt.xlabel('X Position')
plt.ylabel('Y Position')
plt.title('Active Brownian Particle Positions \n DataSet1 initial')
plt.show()
```

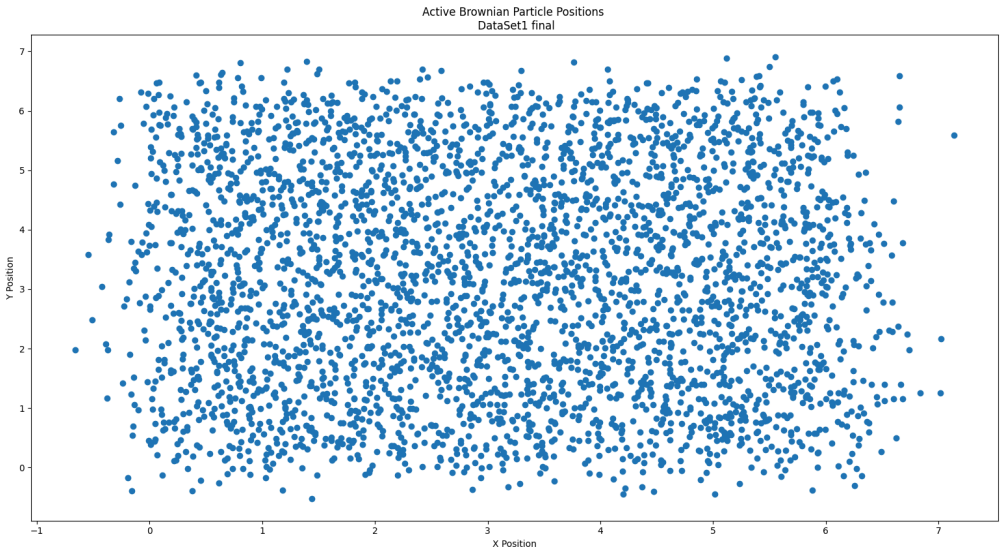
Graph:



Final Position

```
import numpy as np
import matplotlib.pyplot as plt

file_path = r'C:\Users\LC15\Desktop\Stat
Mech\Active_BrownianParticle_DataSet\Active_BrownianParticle_DataSet\Final
ized_data\DataSet1\part\POSITION10000'
with open(file_path, 'rb') as myFile:
    data = np.fromfile(myFile, dtype=np.float64)

x_position = data[0:3517]
y_position = data[3517:7035]

print(data)
print(len(data))
plt.scatter(x_position, y_position)
plt.xlabel('X Position')
plt.ylabel('Y Position')
plt.title('Active Brownian Particle Positions \n DataSet1 final')
plt.show()
```

Graph:

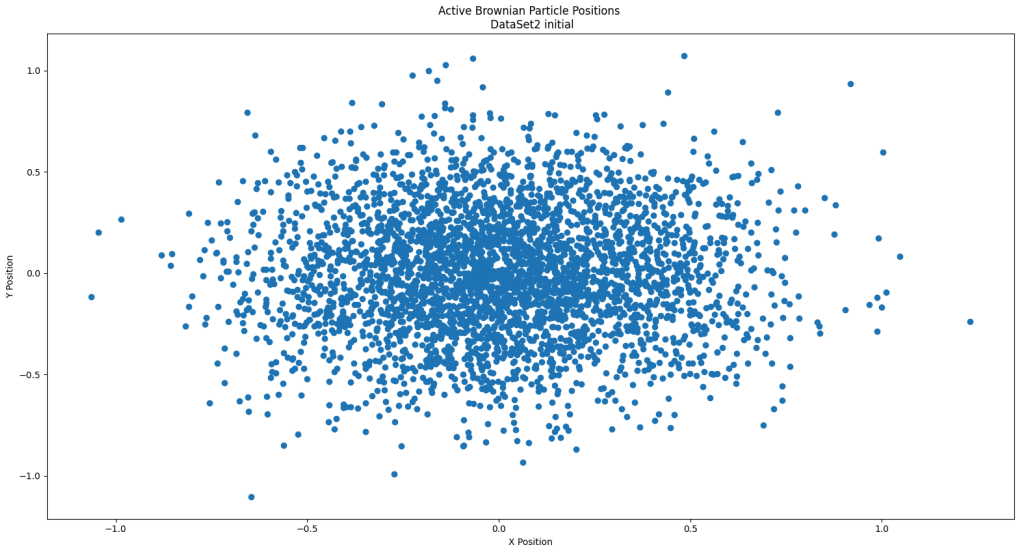


Similarly I have done it for DataSet2 , DataSet3 and DataSet4

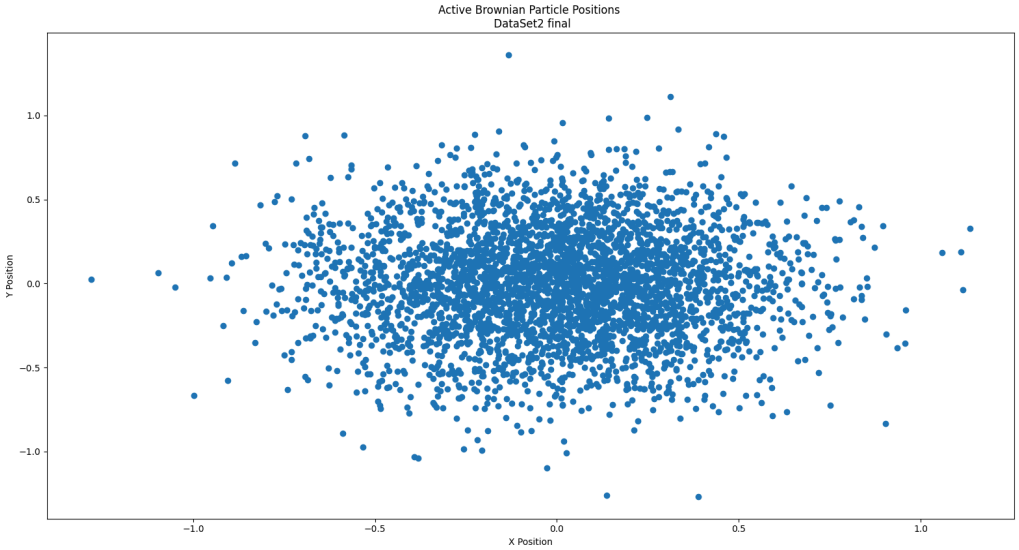
For DataSet2 and DataSet3 , we have to change only the file path to plot the graph for initial and final position.while as for DataSet4 we have to change some other things like number of particles.

DataSet2

Initial position Graph:

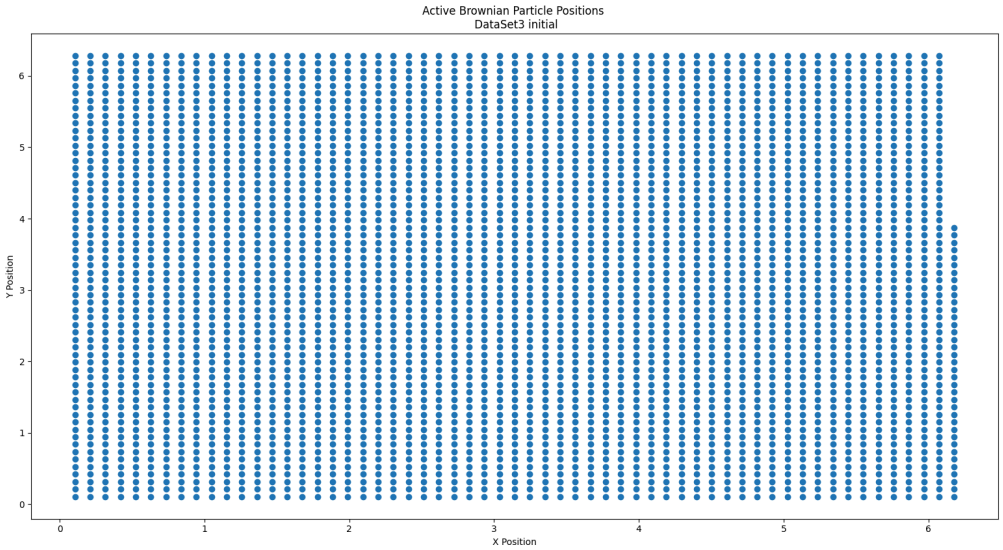


Final position Graph:

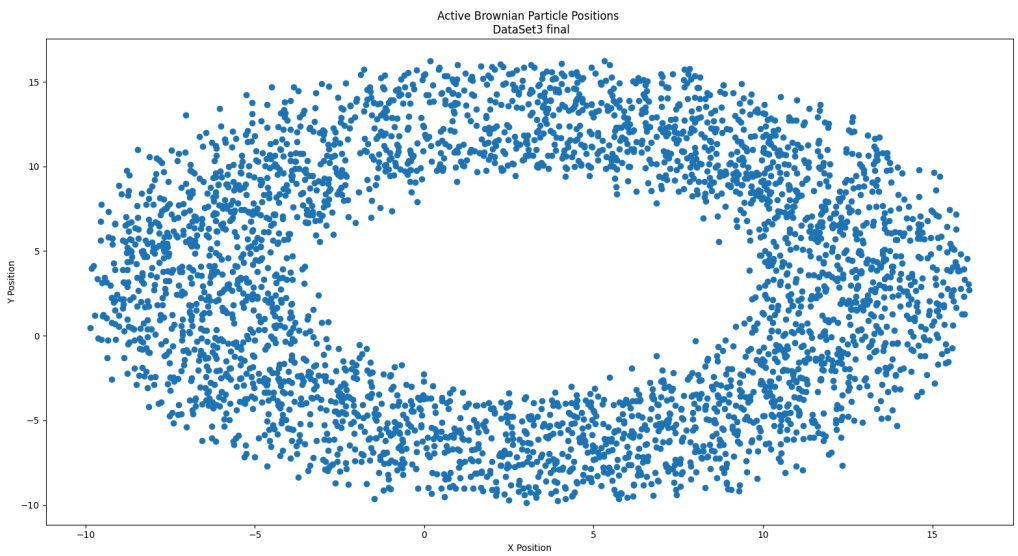


DataSet3

Initial position Graph:



Final position Graph:

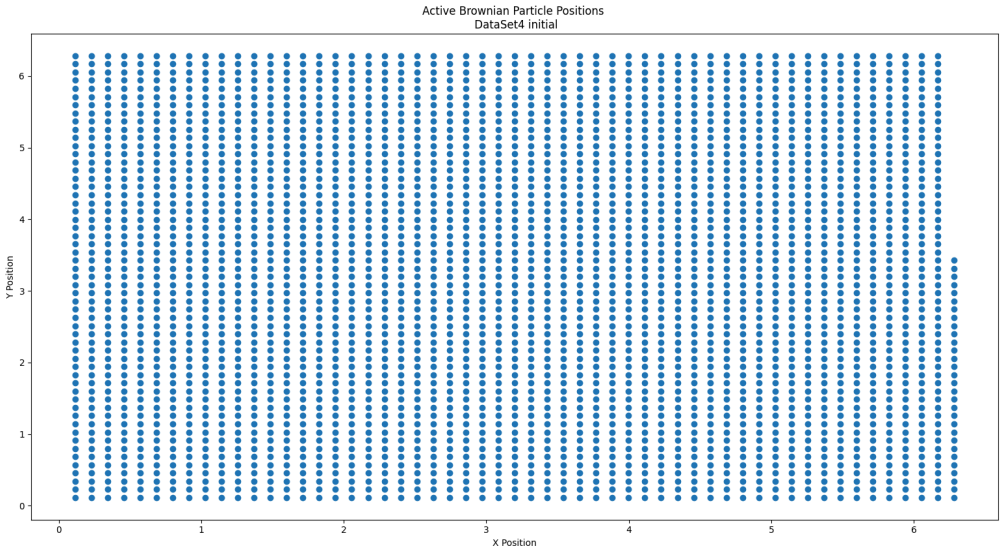


DataSet4 has the 1000000 files and each file contains N = 3000 particles

Initial position

```
import numpy as np
import matplotlib.pyplot as plt

file_path = r'C:\Users\LC15\Desktop\Stat
Mech\Active_BrownianParticle_DataSet\Active_BrownianParticle_DataSet\Final
ized_data\DataSet4\part\POSITION0'
with open(file_path, 'rb') as myFile:
    data = np.fromfile(myFile, dtype=np.float64)

x_position = data[0:3000]
y_position = data[3000:6001]

print(data)
print(len(data))
plt.scatter(x_position, y_position)
plt.xlabel('X Position')
plt.ylabel('Y Position')
plt.title('Active Brownian Particle Positions \n DataSet4 initial')
plt.show()
```

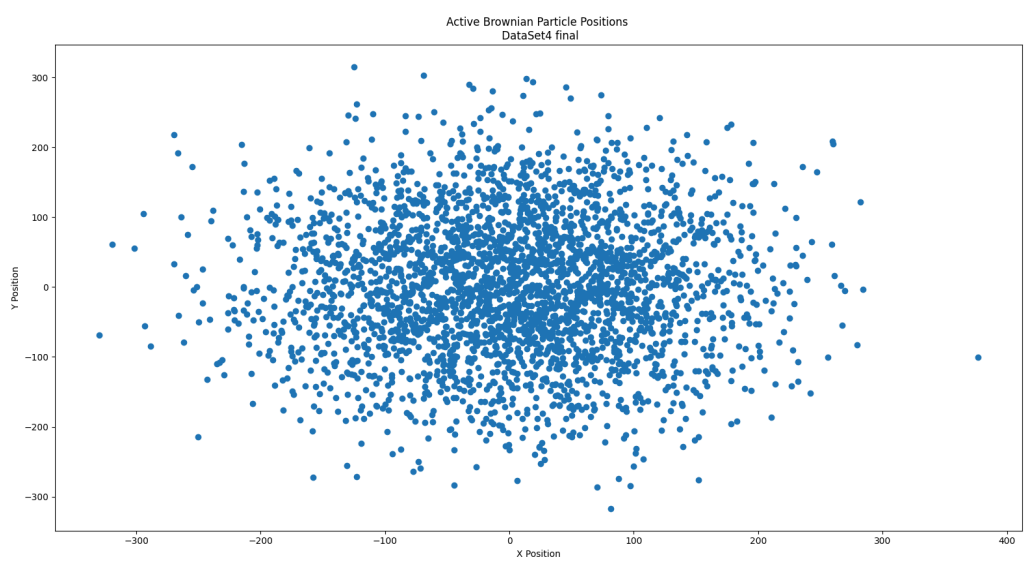
Graph:



Final position

```
import numpy as np
import matplotlib.pyplot as plt

file_path = r'C:\Users\LC15\Desktop\Stat
Mech\Active_BrownianParticle_DataSet\Active_BrownianParticle_DataSet\Final
ized_data\DataSet4\part\POSITION1000000'
with open(file_path, 'rb') as myFile:
    data = np.fromfile(myFile, dtype=np.float64)

x_position = data[0:3000]
y_position = data[3000:6001]

print(data)
print(len(data))
plt.scatter(x_position, y_position)
plt.xlabel('X Position')
plt.ylabel('Y Position')
plt.title('Active Brownian Particle Positions \n DataSet4 final')
plt.show()
```

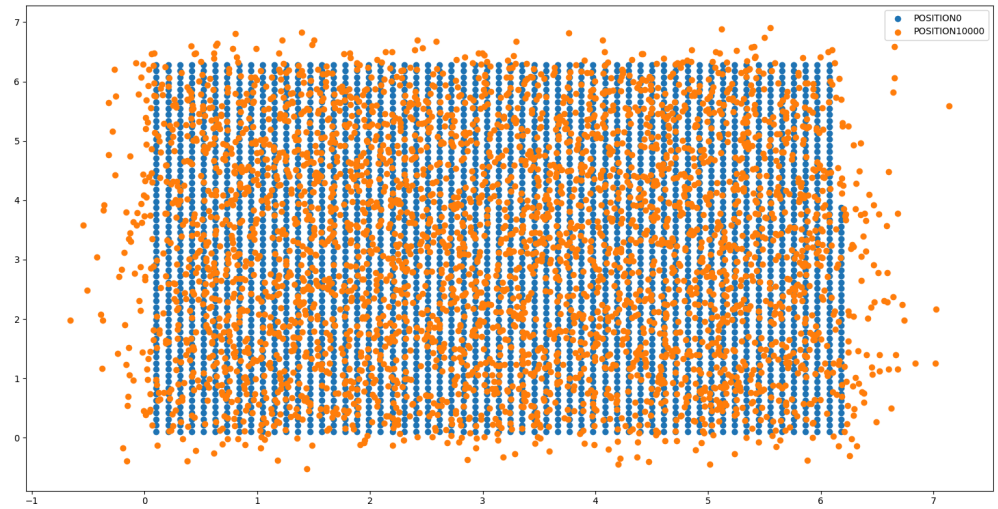
Graph:



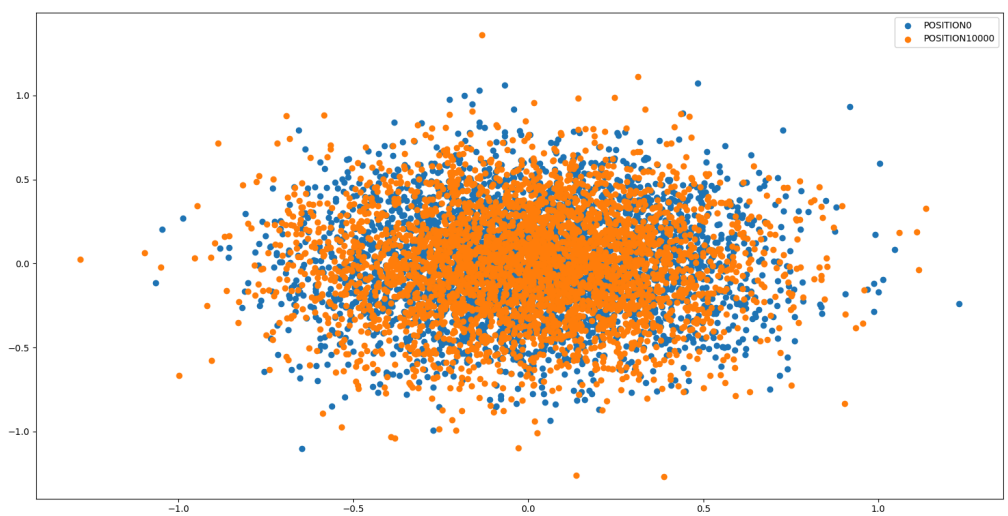
If we want to analyze the initial and final positions of the particles in one graph then we can use the following code which helps us to analyze the particle data using different colors for the initial and final positions.

```
import numpy as np
import matplotlib.pyplot as plt

for i in [0, 10000]:
    file_path = rf'C:\Users\LC15\Desktop\Stat
Mech\Active_BrownianParticle_DataSet\Active_BrownianParticle_DataSet\Final
ized_data\DataSet1\part\POSITION{i}'
    data = np.fromfile(file_path, dtype=np.float64)
    x_position = data[0:3517]
    y_position = data[3517:7035]
    plt.scatter(x_position, y_position, label=f'POSITION{i}')
print(data)
print(len(data))
plt.xlabel('X Position')
plt.ylabel('Y Position')
plt.title('Active Brownian Particle Positions')
plt.legend()
plt.show()
```

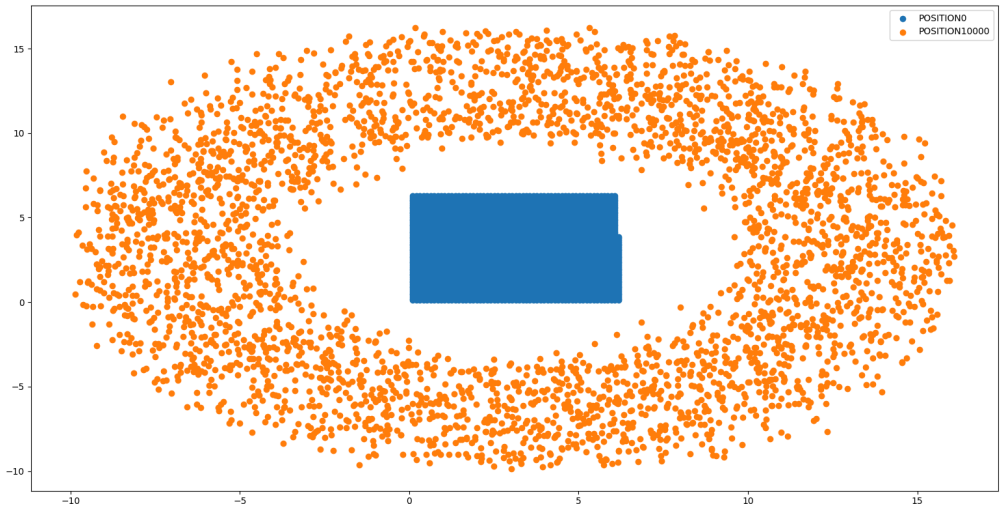
For data set 1



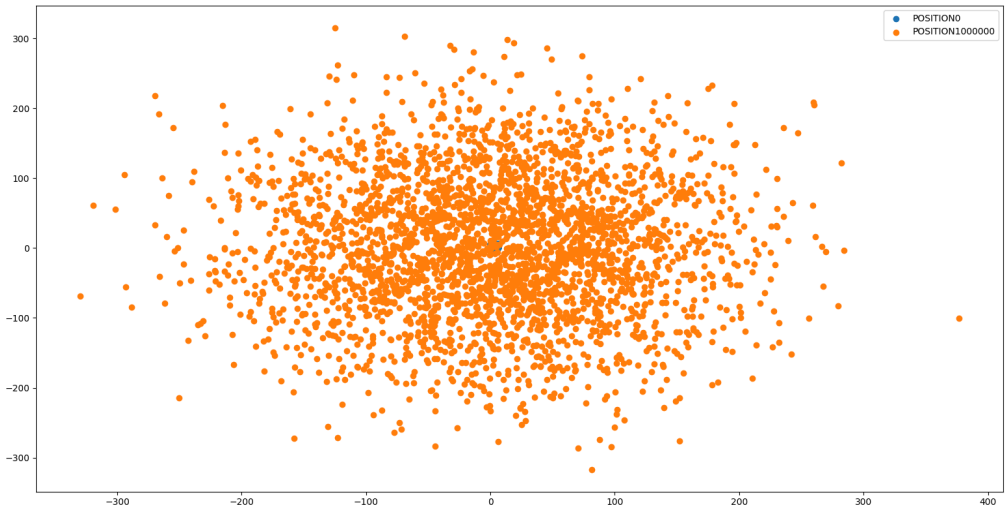
For data set 2



For data set 3



For data set 4

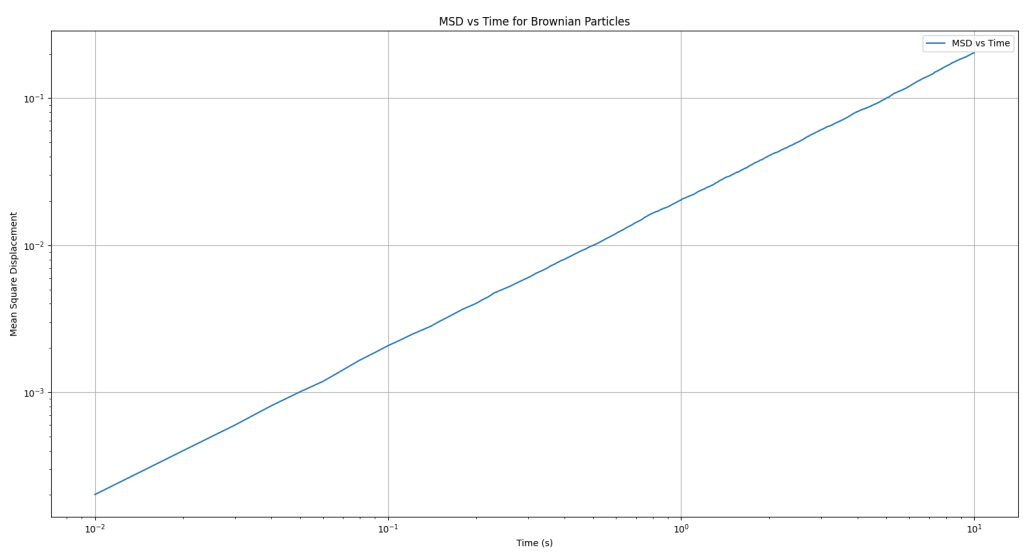


_____PART B_____

For data set 1

```
import numpy as np
import matplotlib.pyplot as plt

# Load initial positions (snapshot 0)
file_path_initial = r'C:\Users\LC15\Desktop\Stat
Mech\Active_BrownianParticle_DataSet\Active_BrownianParticle_DataSet\Finalized_da
ta\DataSet1\part\POSITION0'
with open(file_path_initial, 'rb') as myinitialFile:
    initial_data = np.fromfile(myinitialFile, dtype=np.float64)

N = 3517
x_position_i = initial_data[0:N]
y_position_i = initial_data[N:2*N]

times = []
MSD_values = []

# Loop through snapshots from 10 to 10000 in steps of 10
for i in range(10, 10001, 10):
    file_path = rf'C:\Users\LC15\Desktop\Stat
Mech\Active_BrownianParticle_DataSet\Active_BrownianParticle_DataSet\Finalized_da
ta\DataSet1\part\POSITION{i}'
    with open(file_path, 'rb') as f:
        data = np.fromfile(f, dtype=np.float64)

    x_position_f = data[0:N]
    y_position_f = data[N:2*N]

    # Calculate displacement differences
    x_diff = x_position_f - x_position_i
    y_diff = y_position_f - y_position_i

    # Calculate squared displacement and mean over all particles
    diff_sq_sum = np.square(x_diff) + np.square(y_diff)
```

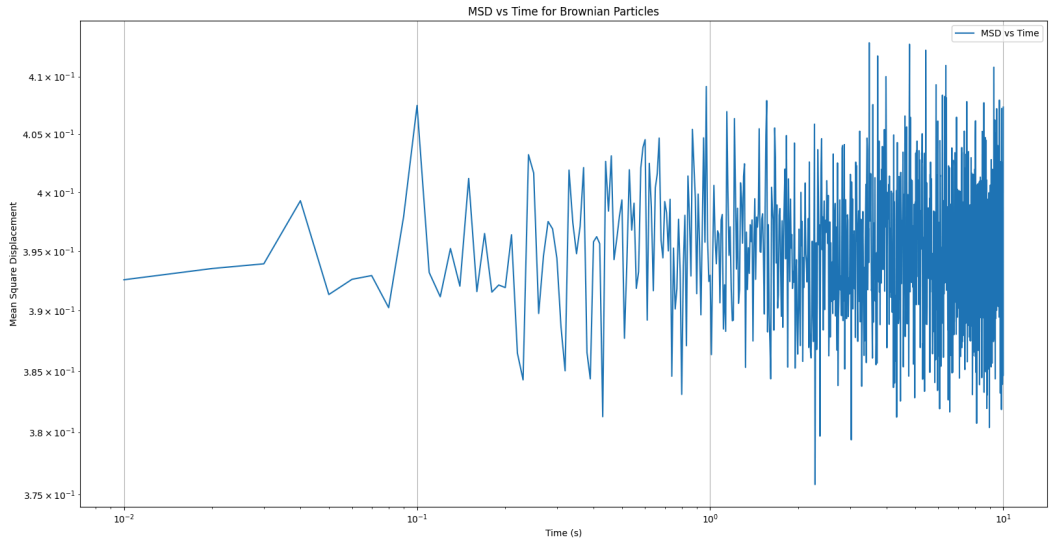
```
    MSD = np.mean(diff_sq_sum)

    time_diff = i * 0.001  # 0.001s between snapshots times snapshot index
    times.append(time_diff)
    MSD_values.append(MSD)


# Plot graph of MSD vs time
plt.loglog(times, MSD_values, label='MSD vs Time')
plt.xlabel('Time (s)')
plt.ylabel('Mean Square Displacement')
plt.title('MSD vs Time for Brownian Particles')
plt.legend()
plt.grid(True)
plt.show()
```



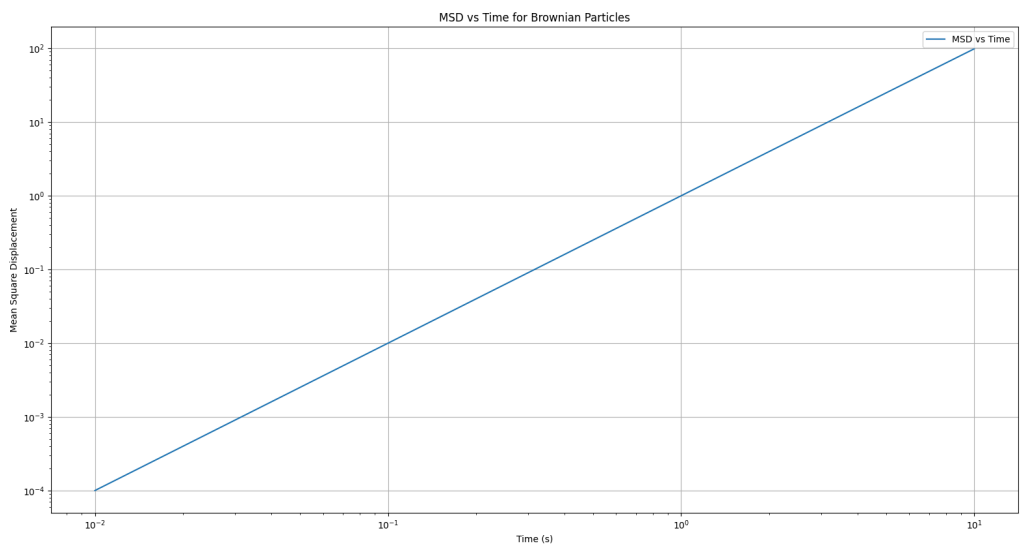
Similarly I have done it for DataSet2 , DataSet3 and DataSet4

For DataSet2 , DataSet3 we have to change only the file path.

For data set 2



For data set 3

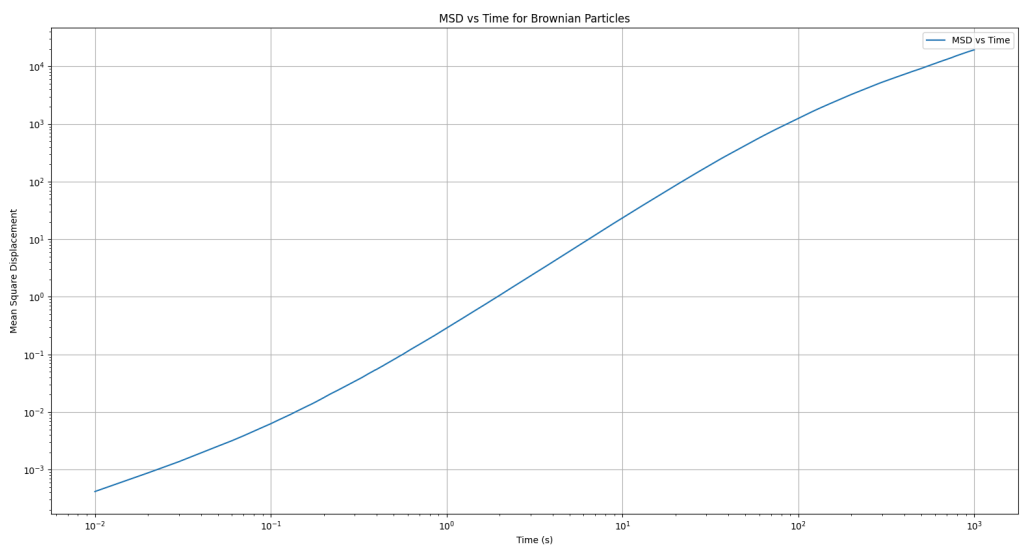


For data set 4

```
# for dataset4
```

```
import numpy as np
import matplotlib.pyplot as plt

# Load initial positions (snapshot 0)
file_path_initial = r'C:\Users\LC15\Desktop\Stat
Mech\Active_BrownianParticle_DataSet\Active_BrownianParticle_DataSet\Final
ized_data\DataSet4\part\POSITION0'
with open(file_path_initial, 'rb') as myinitialFile:
    initial_data = np.fromfile(myinitialFile, dtype=np.float64)

N = 3000
x_position_i = initial_data[0:N]
y_position_i = initial_data[N:2*N]

times = []
MSD_values = []

# Loop through snapshots from 10 to 10000 in steps of 10
for i in range(10, 1000001, 10):
    file_path = rf'C:\Users\LC15\Desktop\Stat
Mech\Active_BrownianParticle_DataSet\Active_BrownianParticle_DataSet\Final
ized_data\DataSet4\part\POSITION{i}'
    with open(file_path, 'rb') as f:
        data = np.fromfile(f, dtype=np.float64)

    x_position_f = data[0:N]
    y_position_f = data[N:2*N]

    # Calculate displacement differences
    x_diff = x_position_f - x_position_i
    y_diff = y_position_f - y_position_i

    # Calculate squared displacement and mean over all particles
    diff_sq_sum = np.square(x_diff) + np.square(y_diff)
    MSD = np.mean(diff_sq_sum)

    time_diff = i * 0.001  # 0.001s between snapshots times snapshot index
    times.append(time_diff)
    MSD_values.append(MSD)
```

```
# Plot graph of MSD vs time
plt.loglog(times, MSD_values, label='MSD vs Time')
plt.xlabel('Time (s)')
plt.ylabel('Mean Square Displacement')
plt.title('MSD vs Time for Brownian Particles')
plt.legend()
plt.grid(True)
plt.show()
```



Google Drive link for code & better view of graphs:

Stat Mech Brownian Motion

I really enjoyed doing this assignment:)